CAMPUS EVENT FINANCIAL PLANNING

Project Report — Python Programming

Project purpose	Estimate event costs, participant contributions, sponsorship gap, and break-even registrations
Cost categories	Venue, equipment, catering, transport, and certificates
Calculations	Total budget, per-participant variable cost, sponsor shortfall, break-even count
Implementation	Python 3, console-based program

All amounts and sample values are illustrative. Adjust them to match the event's actual quotations and policy.

1. Problem Statement

Planning a campus event requires estimating both fixed costs and costs that increase with the number of participants. Common expenses include venue rental, equipment, catering, transport, and certificates. Organizers also need to estimate participant registration fees and sponsorship income, identify any funding shortfall, and calculate the minimum number of registrations needed to cover costs. This project builds a Python program to perform these calculations and present a clear financial summary.

2. Objectives

• Collect fixed and per-participant costs for venue, equipment, catering, transport, and certificates.

• Calculate the total fixed cost and variable cost per participant.

• Estimate total event cost for a planned registration count.

• Calculate income from participant contributions and sponsorship.

• Identify whether a funding shortfall or surplus remains.

• Calculate the minimum registration count needed to break even, where mathematically possible.

3. Assumptions

• Venue and equipment are treated as fixed costs for the event.

• Catering, transport, and certificates are treated as per-participant costs.

• Every registered participant pays the same registration contribution.

• Sponsorship is a fixed confirmed amount and is available to cover event costs.

• No refunds, taxes, contingency costs, or other revenue sources are included.

• Break-even registration count uses the contribution per participant minus variable cost per participant.

• If participant contribution is less than or equal to variable cost per participant, registrations alone cannot cover fixed costs; the program reports that break-even is not possible under the current rates.

4. Algorithm

1. Input fixed costs for venue and equipment.

2. Input per-participant costs for catering, transport, and certificates.

3. Input planned registration count, participant contribution per registration, and confirmed sponsorship.

4. Calculate total fixed cost and variable cost per participant.

5. Calculate projected total cost = fixed cost + variable cost per participant × planned registrations.

6. Calculate participant revenue = registration count × contribution per participant.

7. Calculate total funding = participant revenue + sponsorship.

8. Calculate surplus or shortfall = total funding – projected total cost.

9. Calculate uncovered fixed cost after sponsorship and divide by the positive contribution margin to find the minimum break-even registrations; round up to a whole participant.

10. Display all cost components, funding, shortfall/surplus, and break-even result.

5. Flowchart

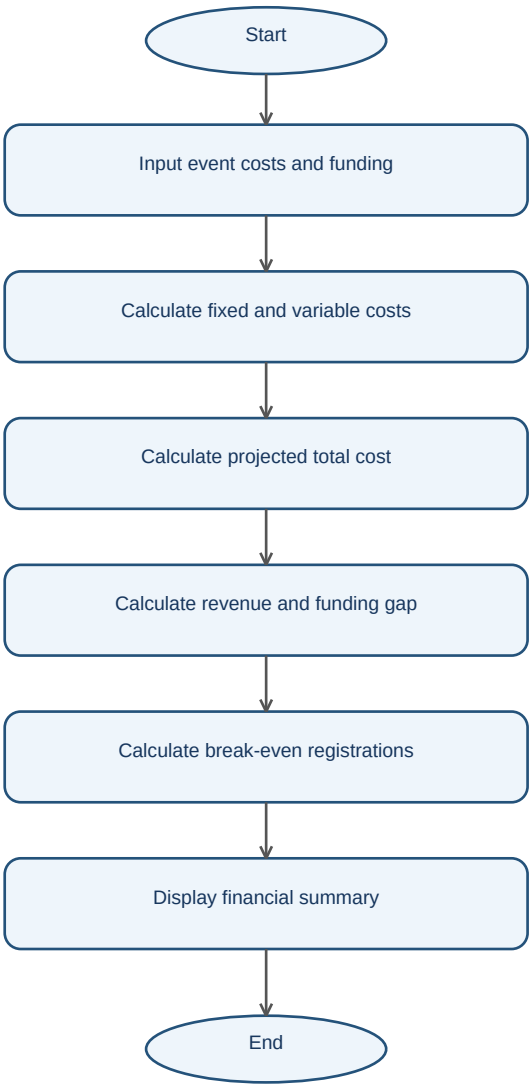

Figure 1. High-level flow of the campus event financial planning program.

6. Pseudocode

```
START
  INPUT venue_cost, equipment_cost
  INPUT catering_per_person, transport_per_person, certificate_per_person
  INPUT planned_registrations, fee_per_person, sponsorship

  fixed_cost = venue_cost + equipment_cost
  variable_cost = catering_per_person + transport_per_person
                + certificate_per_person
  projected_cost = fixed_cost + variable_cost * planned_registrations
  participant_revenue = fee_per_person * planned_registrations
  total_funding = participant_revenue + sponsorship
  balance = total_funding - projected_cost

  contribution_margin = fee_per_person - variable_cost
  uncovered_fixed_cost = max(0, fixed_cost - sponsorship)
  IF contribution_margin > 0
    break_even = CEILING(uncovered_fixed_cost / contribution_margin)
  ELSE
    break_even = "Not possible at current contribution rate"
  END IF

  DISPLAY cost breakdown, funding, balance, break_even
END
```

7. Python Source Code

```
import math

def read_money(prompt):
    while True:
        try:
            value = float(input(prompt))
            if value < 0:
                print("Amount cannot be negative.")
            else:
                return value
        except ValueError:
            print("Enter a valid amount.")

def read_int(prompt, minimum=0):
    while True:
        try:
            value = int(input(prompt))
            if value < minimum:
                print(f"Enter a whole number at least {minimum}.")
            else:
                return value
        except ValueError:
            print("Enter a valid whole number.")

def main():
    print("CAMPUS EVENT FINANCIAL PLANNING")
    venue = read_money("Venue rental cost (Rs): ")
    equipment = read_money("Equipment cost (Rs): ")
    catering = read_money("Catering cost per participant (Rs): ")
    transport = read_money("Transport cost per participant (Rs): ")
    certificate = read_money("Certificate cost per participant (Rs): ")
    registrations = read_int("Planned number of registrations: ", 0)
    fee = read_money("Participant contribution per registration (Rs): ")
    sponsorship = read_money("Confirmed sponsorship (Rs): ")

    fixed_cost = venue + equipment
    variable_per_person = catering + transport + certificate
    projected_cost = fixed_cost + variable_per_person * registrations
    participant_revenue = fee * registrations
    total_funding = participant_revenue + sponsorship
    balance = total_funding - projected_cost

    contribution_margin = fee - variable_per_person
    uncovered_fixed = max(0, fixed_cost - sponsorship)

    print("\\nFINANCIAL SUMMARY")
    print(f"Fixed costs (venue + equipment): Rs {fixed_cost:.2f}")
    print(f"Variable cost per participant:   Rs {variable_per_person:.2f}")
    print(f"Projected total event cost:      Rs {projected_cost:.2f}")
    print(f"Participant contributions:       Rs {participant_revenue:.2f}")
    print(f"Sponsorship:                     Rs {sponsorship:.2f}")
    print(f"Total funding:                   Rs {total_funding:.2f}")

    if balance < 0:
        print(f"Funding shortfall:               Rs {-balance:.2f}")
        print("Consider increasing sponsorship, registration fee, or registrations.")
    elif balance > 0:
        print(f"Projected surplus:               Rs {balance:.2f}")
    else:
        print("Funding status: exactly break-even.")
```

```
    if contribution_margin > 0:
        break_even = math.ceil(uncovered_fixed / contribution_margin)
        print(f"Break-even registrations: {break_even}")
        print("This is the minimum count under the entered cost and funding assumptions.")
    else:
        print("Break-even registrations: not possible at the current fee.")
        print("The fee must exceed variable cost per participant to cover fixed costs.")

if __name__ == "__main__":
    main()
```

8. Test Cases

Test	Sample scenario	Expected result
TC-01	Fixed ■20,000; variable ■250/person; 100 registrations; fee ■400; sponsor ■10,000	Cost ■45,000; funding ■50,000; surplus ■5,000; break-even 40 registrations
TC-02	Fixed ■30,000; variable ■300/person; 50 registrations; fee ■350; sponsor ■5,000	Cost ■45,000; funding ■22,500; shortfall ■22,500; break-even 500 registrations
TC-03	Fee equals variable cost per participant	Report break-even as not possible at current fee
TC-04	Zero registrations and zero sponsorship	Projected cost equals fixed cost; funding shortfall equals fixed cost
TC-05	Negative cost entered	Program rejects the value and asks again
TC-06	Invalid text entered for registrations	Program rejects it and requests a whole number

9. Sample Output / Console Screenshot

Sample inputs: venue ■15,000; equipment ■5,000; catering ■150/person; transport ■60/person; certificate ■40/person; 100 registrations; contribution ■400/person; sponsorship ■10,000.

```
CAMPUS EVENT FINANCIAL PLANNING
Venue rental cost (Rs): 15000
Equipment cost (Rs): 5000
Catering cost per participant (Rs): 150
Transport cost per participant (Rs): 60
Certificate cost per participant (Rs): 40
Planned number of registrations: 100
Participant contribution per registration (Rs): 400
Confirmed sponsorship (Rs): 10000

FINANCIAL SUMMARY
Fixed costs (venue + equipment): Rs 20000.00
Variable cost per participant:   Rs 250.00
Projected total event cost:      Rs 45000.00
Participant contributions:       Rs 40000.00
Sponsorship:                     Rs 10000.00
Total funding:                   Rs 50000.00
Projected surplus:                Rs 5000.00
Break-even registrations: 40
```

This is an illustrative console transcript, not an actual screenshot captured from a running program. For actual screenshots, run the program in IDLE or VS Code, enter the sample values, and capture the code/output window using Win + Shift + S on Windows.

10. Results and Discussion

The program separates fixed event costs from per-participant expenses. It estimates the total cost for the planned attendance, compares available funding against that cost, and reports the shortfall or surplus. Break-even is calculated using the contribution margin after variable cost. In this simplified model, sponsorship offsets fixed costs first; if the registration fee does not exceed variable cost per participant, registrations cannot recover remaining fixed costs.

11. Conclusion

The Campus Event Financial Planning project demonstrates how Python can support budgeting, cost classification, revenue calculation, and break-even analysis. It helps organizers estimate the funding needed before confirming an event and understand how registration fees, sponsorship, and attendance affect the budget. For real planning, organizers should add taxes, contingency reserves, refunds, tiered fees, and verified supplier quotations.

12. How to Run

• Save the code in a file named campus_event_finance.py.

• Run it using Python 3: python campus_event_finance.py.

• Enter event costs, planned registrations, participant contribution, and confirmed sponsorship.

• Compare the financial summary with the test cases.